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Introduction to AI Agents on Azure

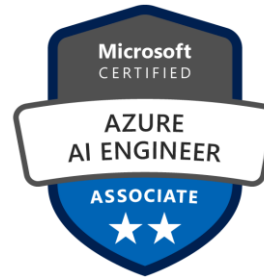
Learn to Develop, Test, and Deploy AI Agents
with Microsoft Foundry



Reza Salehi

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Microsoft Azure Fundamentals (AZ-900) Certification Course, 2nd Edition



4h 55m remaining

With your instructor

[Reza Salehi](#)

[+ Add to playlist](#)

Associated roles

Cloud native engineer

Cloud solutions architect

Cybersecurity engineer

Database administrator

[+1 more](#)

Skills covered

AZ-900: Microsoft Azure Fundamentals

AZ-303: Microsoft Azure Architect...

AZ-500: Microsoft Azure Security...

AI-900: Microsoft Azure AI Fundamentals



Includes quizzes

Test your knowledge during the course and with a final quiz.

 Beginner

Instructional content for a learner who has little or no knowledge on this topic.

Learning Outcomes

- Gain knowledge of Azure cloud concepts and services
- Explore Azure services in greater depth
- Get ready for Exam AZ-900: Microsoft Azure Fundamentals
- Comfortably work with the Azure portal

The Microsoft Azure Fundamentals (AZ-900) exam is one of the most popular certifications for those who are just beginning to work with cloud-based solutions and services or who are new to Azure. The exam certifies knowledge of cloud concepts, Azure services, workloads, security and privacy, and pricing and support.

In this self-paced course, Reza Salehi will help you get familiar with Microsoft Azure's cloud services and begin your Azure certification journey. This course is aligned to the AZ-900 exam objective domains and has recently been updated to reflect the most current version of the exam (2024). It covers all the services and concepts in the Azure ecosystem you need to know in order to prepare for the test.

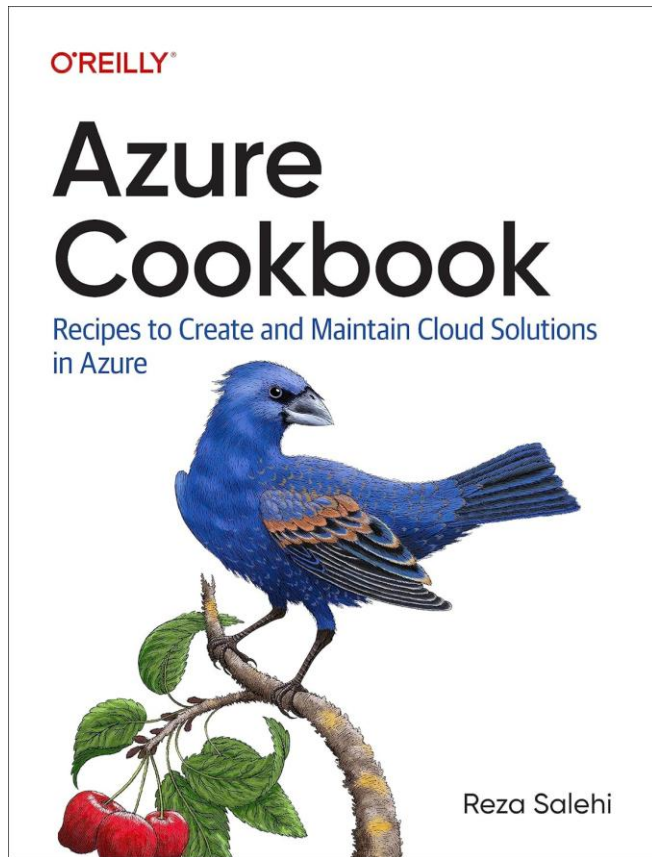
What you'll learn and how to apply it

By the end of this certification course, you will understand the following:

- General cloud concepts
- Core Azure services
- Core solutions and management tools on Azure
- General security and network security features
- Identity, governance, privacy, and compliance features
- Azure cost management and service-level agreements

Azure Cookbook

- <https://learning.oreilly.com/library/view/azure-cookbook/9781098135782/>
- <https://www.amazon.ca/Azure-Cookbook-Recipes-Maintain-Solutions/dp/1098135792/>
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Course Overview





Learning Objectives

By the end of this course, you will:

- Gaining familiarity with Microsoft Foundry
- Use a variety of AI models for your AI agent
- Leverage Microsoft Foundry for agent development





Course Repository

<https://github.com/zaalion/oreilly-ai-agents-designing>





CERTIFICATION

Microsoft Certified: Azure AI Fundamentals

Demonstrate fundamental AI concepts related to the development of software and services of Microsoft Azure to create AI solutions.

At a glance

 **Level**
Beginner

 **Product**
Azure

 **Role**

 **Subject**



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Design and implement an Azure AI solution using Azure AI services, Azure AI Search, and Azure Open AI.

At a glance

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Intermediate

 **Role**

 **Product**
Azure

 **Subject**

From cloud to agents:
Discover the next
generation of Microsoft
Certifications





CERTIFICATION

Microsoft Certified: Agentic AI Business Solutions Architect

As an AI-first solution architect, you lead the transformation of enterprise operations by envisioning and implementing AI-powered architecture.

At a glance

 **Level**
Advanced

 **Role**
Solution Architect

 **Renewal Frequency**
12 months

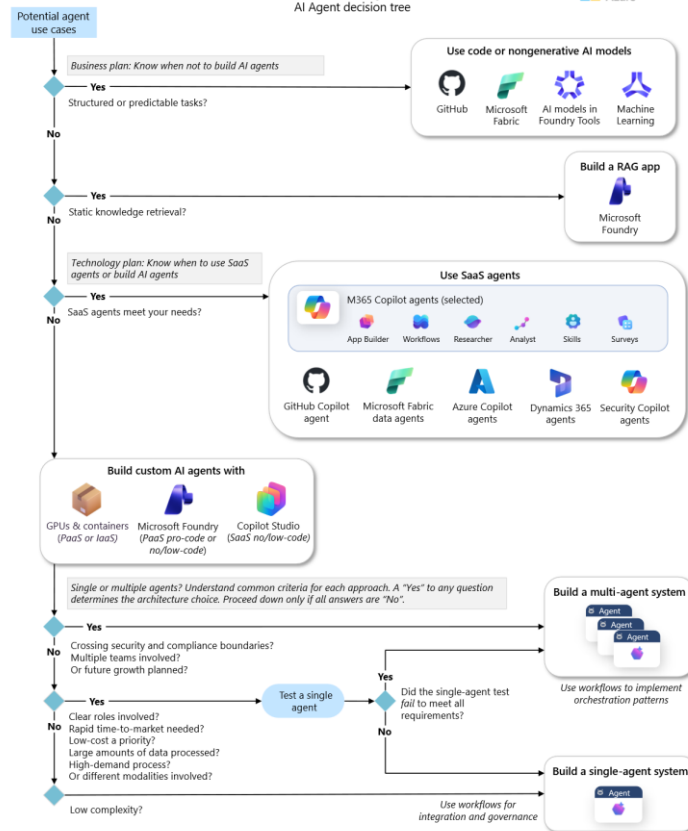
 **Product**
AI Builder

 **Subject**
Artificial intelligence

Technology Plan For AI Agents



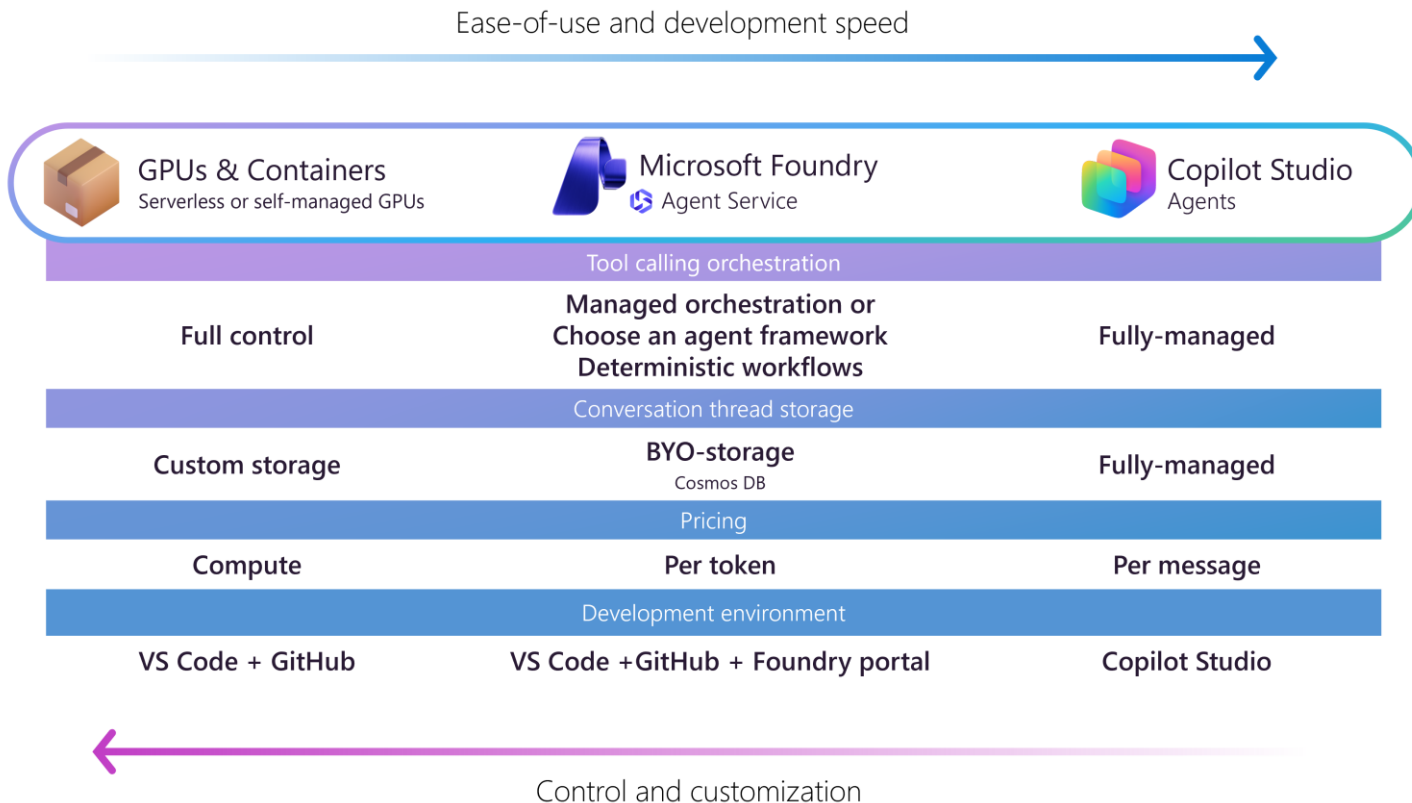
Microsoft's Cloud Adoption Framework AI Agent decision tree



<https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ai-agents/technology-solutions-plan-strategy>



Technology Plan For AI Agents





Pulse check: Have you developed an AI agent?





Understanding LLMs in Microsoft Foundry





Azure OpenAI in Azure AI Foundry Models

- Azure OpenAI Service
 - Available models
 - Pricing
 - Quotas & limits
 - Deploy, and use a model
 - Upgrade to Microsoft Foundry
- Microsoft Foundry (Azure AI Foundry)
 - Model catalog
 - Agent Playground
 - Instructions, actions, triggers, tools, RAG, MCP



Poll: What is Azure OpenAI Service?

- A marketplace for third-party AI apps
- New name for Azure Machine Learning
- A database service for storing AI data
- A managed service to access OpenAI models in Azure



Azure OpenAI in Azure AI Foundry

1. Azure OpenAI

- a. Provides REST API access to OpenAI's powerful language models including gpt-5 series, o4-mini, and more.

2. Microsoft Foundry

- a. Unifies agents, models, and tools under a single management grouping with tracing, monitoring, evaluating, and deployment capabilities.



Azure OpenAI Models

- [Models available](#)
 - gpt-5 series
 - o4-mini & o3
 - gpt-4.1
 - computer-use-preview
 - o3-mini & o1
 - o1-mini
 - GPT-4o & GPT-4o mini
 - GPT-4 series (including GPT-4 Turbo with Vision)
 - GPT-3.5-Turbo series



Azure OpenAI vs. OpenAI

- OpenAI + Azure = Azure OpenAI
 - Native integration with other Azure resources
 - Rich tooling (Azure AI Foundry, Visual Studio, VS Code, etc.)
- Azure OpenAI co-develops APIs and models with OpenAI
- Is compatible and provides smooth transition from OpenAI.



Azure OpenAI Pricing: Pay Per Token

GPT-5 series

Model	Pricing (1M Tokens)
GPT-5 2025-08-07 Global	Input: \$1.25 Cached Input: \$0.13 Output: \$10
GPT-5 Data Zone	Input: \$1.38 Cached Input: \$0.14 Output: \$11
GPT-5-mini Global	Input: \$0.25 Cached Input: \$0.03 Output: \$2
GPT-5-mini Data Zone	Input: \$0.28 Cached Input: \$0.03 Output: \$2.20

<https://azure.microsoft.com/en-us/pricing/details/cognitive-services/openai-service/>



Azure OpenAI Pricing: Pay Per Token

 | **Azure** Explore Products Solutions Pricing Partners Resources

Search Learn Support Contact Sales [Get started with Azure](#) [Sign in](#)

Pricing calculator

Calculate your estimated hourly or monthly costs for using Azure.

[Get started with Azure](#)[Log in](#) to save cost estimates and use your Azure agreement pricing.

Products

Example scenarios

Saved estimates

FAQs

Select a product to include it in your estimate.

openai

 **Azure OpenAI** ⓘ

Apply advanced coding and language models to a variety of use cases

[Add to estimate](#)

<https://azure.microsoft.com/en-us/pricing/calculator>



Azure OpenAI Pricing: Token Types

- Tokens
 - [Text tokens](#)
 - Azure OpenAI processes text by breaking it into tokens.
 - Can be words or chunks of characters (ham.bur.ger, pear, etc.)
 - [Image input tokens](#)
 - Level of image detail (low or high) + the image dimensions
 - [Image generation tokens](#)
 - Image dimensions and quality

<https://learn.microsoft.com/en-us/azure/ai-foundry/openai/overview#tokens>



Poll: What is Microsoft Foundry?

- A tool for labeling AI training data
- A platform to build, customize, and deploy generative AI apps
- A service for hosting machine learning notebooks
- A marketplace for pre-trained AI models



Microsoft Foundry

- As a developer, [Microsoft Foundry](#) helps you:
 - Explore, build, test, and deploy generative AI models
 - Grounded in responsible AI practices.
 - Build generative AI applications and AI agents
 - Collaborate with your team for the full development life-cycle
 - Work across model providers with a consistent API interface
 - OpenAI and other 3rd party models



Microsoft Foundry

- Microsoft Foundry [API and SDKs](#)
 - Python
 - C#
 - JavaScript/TypeScript
 - Java

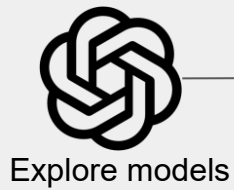
Microsoft Foundry



Microsoft
Foundry Hub



Foundry Project



Build
agent

Test
agent

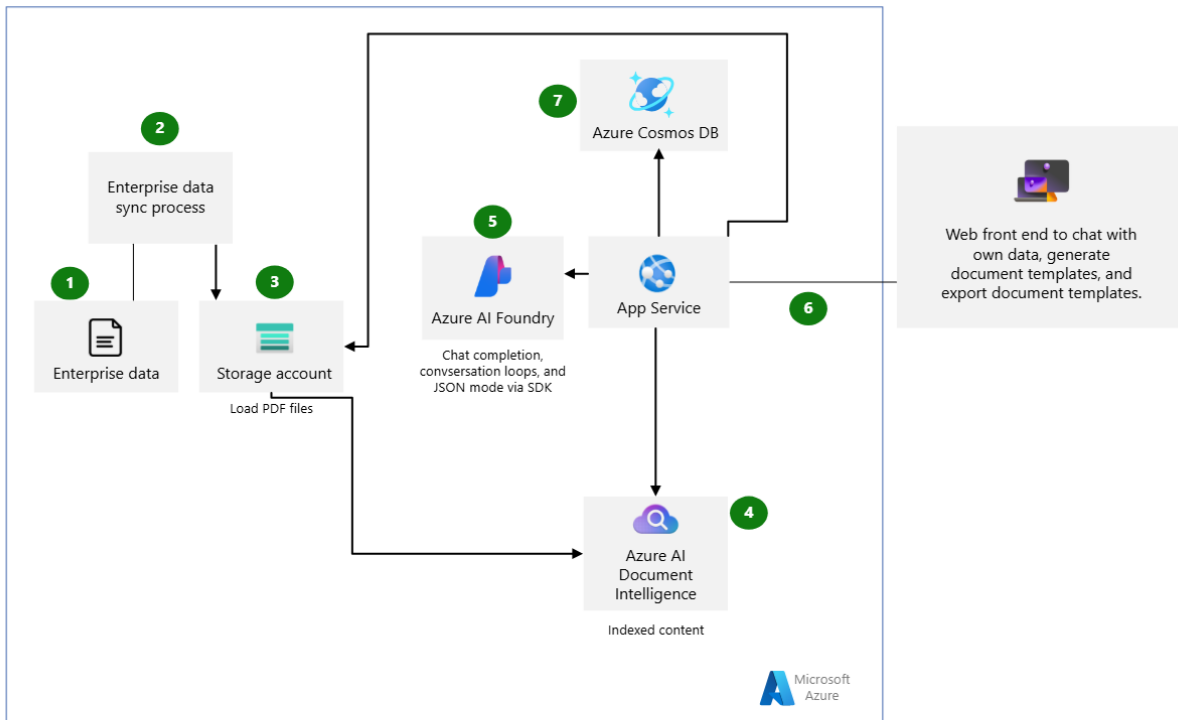
Deploy
agent

Upgrade
(convert)



Azure OpenAI

Microsoft Foundry



<https://learn.microsoft.com/en-us/azure/architecture/ai-ml/idea/generate-documents-from-your-data>



Upgrade from Azure OpenAI to Microsoft Foundry

- For new projects, start with Foundry
- For existing Azure OpenAI resources, [consider upgrading](#):
 - Microsoft Foundry offers more capabilities compared to the Azure OpenAI
 - Access to a broader model catalog, agents service, and testing capabilities
 - An upgrade option is available
 - Keep existing Azure OpenAI API endpoint, state of work, and security configurations
 - No need to create a new Foundry resource



Azure OpenAI in Microsoft Foundry Models

Quotas and Limits

1. [Regional quota allocation](#)
2. [Quotas and limits reference](#)



Microsoft Foundry Playgrounds

- [Foundry playgrounds:](#)
 - a. AgentOps support for Evaluations and Tracing in the **Agents playground**
 - b. **Images Playground 2.0** for gpt-image-1, Stable Diffusion 3.5 Large, and Bria 2.3 Fast models.
 - c. **Audio playground** for gpt-4o-audio, gpt-4o-transcribe, and gpt-4o-mini-tts models.
 - d. **Video playground** for Azure OpenAI Sora.
 - e. Open in **VS Code** for **Chat and Agents playground**. This feature saves you time by automatically importing your endpoint and key from Foundry to VS Code for multi-lingual code samples.



Demo:

Azure OpenAI



Demo:

Azure OpenAI Upgrade to

Microsoft Foundry



Demo:

Azure OpenAI in Microsoft Foundry



Demo:

Add and Configure Models to Microsoft Foundry



Demo:

Microsoft Foundry Chat Playground



Introduction to AI Agent Development in Microsoft Foundry





Introduction to AI Agent Development in Microsoft Foundry

- Understanding Foundry Agent Service
 - Exploring the Agent Playground
- Creating a new agent
 - Choose the right LLM model
 - Triggers
 - Instructions
 - RAG
 - MCP
 - Tools
 - Threads
 - Actions



Poll: Which generative AI models can't be used by Foundry Agent Service?

- Azure OpenAI LLM models
- All Azure OpenAI models
- Azure OpenAI and other 3rd party models such as DeepSeek
- Azure AI Services models



Developing Agents with Azure AI Foundry

- Foundry Agent Service
 - a. Connects models, tools, and frameworks into the agent.
 - b. From generative model demos to a production-ready product:
 - i. Deploying a LLM model
 - ii. Create and test a new agent
 - iii. Deploy to production
 - iv. Monitor, tune, and improve



Developing Agents with Azure AI Foundry

- AI Agent
 - a. Is not a chatbot!
 - b. Makes decisions, invokes tools, and participates in workflows
 - i. Independently
 - ii. Collaboration
 - 1. Other agents
 - 2. Humans
 - c. Enable AI-powered process automations



Poll: Your Azure AI Agent is using a model trained with data up to September 1, 2023. Can it provide answers about events that happened after that date?

- Yes, but it will hallucinate
- Yes, if you implement RAG
- Yes, it has real-time internet access
- No, unless you upgrade the subscription



Retrieval Augmented Generation (RAG)

1. AI model is the brain of the agent
 - a. Has a cut-off date (e.g. January 15th, 2023)
2. RAG enables your agent to retrieve up-to-date data from various sources:
 - a. Sharepoint
 - b. Files
 - c. Azure AI Search (+ underlying databases)
 - d. Bing Search (web search)
 - e. ...



Model Context Protocol (MCP)

1. AI agents need to access tools and external context
2. [MCP](#) enables your agent to connect to tools and data using a standard protocol:
 - a. Databases
 - b. Files
 - c. Azure AI Search
 - d. SharePoint
 - e. ...



Developing Agents with Microsoft Foundry

- AI Agent components
 - a. **Model (LLM):** Powers reasoning and language understanding
 - i. Azure OpenAI models
 - ii. Other 3rd party models
 - b. **Instructions:** Define the agent's goals, behavior, and constraints
 - c. **Tools:** Let the agent
 - i. Retrieve knowledge (RAG)
 - ii. MCP
 - iii. Take action (send emails, etc.)
 - d. **Trigger**
 - a. Activates an AI agent to start working



Developing Agents with Microsoft Foundry

- Microsoft Foundry is the [assembly line](#) for AI agents, following these steps:
 - a. Select a **model**, as the agent's "brain"
 - b. **Customize** the model to fit your project
 - c. Use **AI tools** to retrieve knowledge (RAG) or take an action
 - d. Use MCP to access knowledge and tools
 - e. Use connected agents for **collaboration/orchestration** scenarios
 - f. **Test and monitor** your agent
 - g. [Foundry](#) applies enterprise-grade **security** features including Entra ID support, RBAC, content filters, encryption, and network isolation

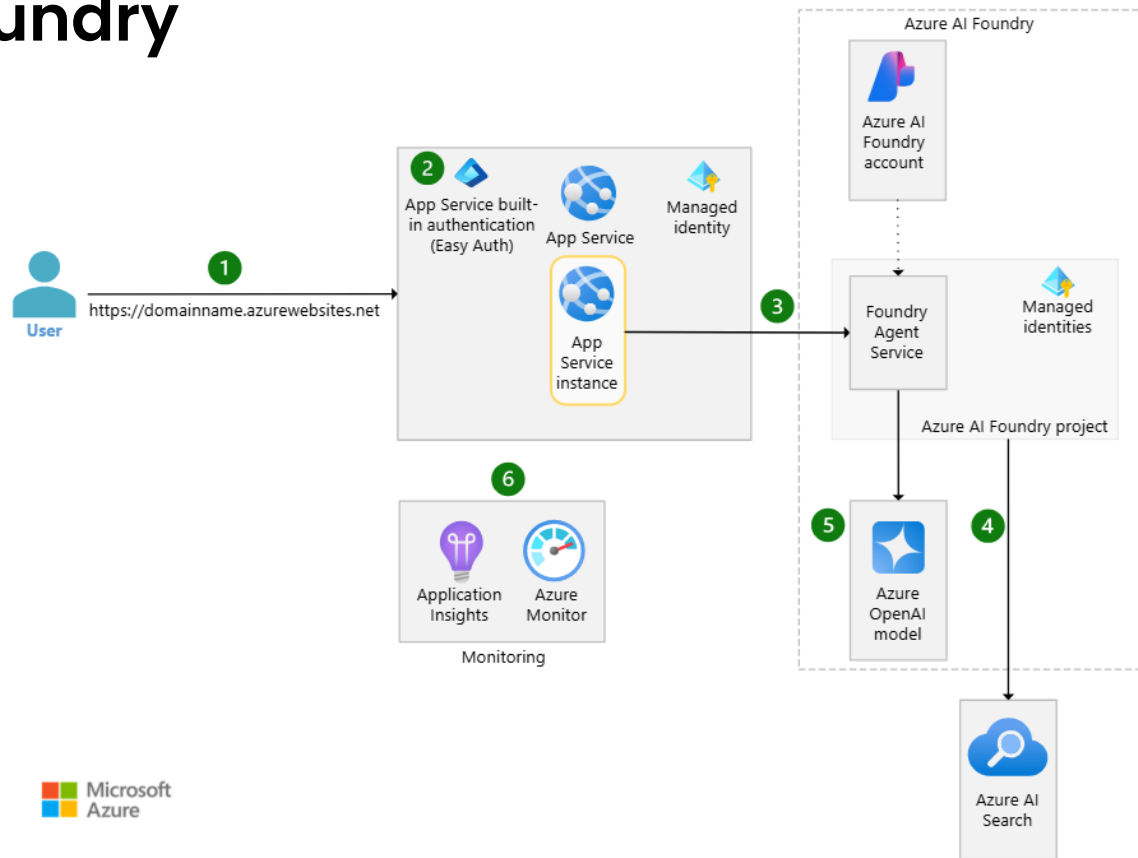


Tools in Microsoft Foundry Agent Service

- Agent tools
 - a. Enables your AI agent to retrieve data or to take actions and automate workflows
 - b. Tool types:
 - i. Knowledge tools
 - 1. Private or public data: Azure AI Search, File Search, Microsoft Fabric, Bing Search, etc.
 - ii. Action tools
 - 1. Azure Logic Apps, Azure Functions, etc.



Microsoft Foundry



<https://learn.microsoft.com/en-us/azure/architecture/ai-ml/architecture/basic-azure-ai-foundry-chat>





Demo:

Create Azure AI Agent Project in

Foundry



Demo:

Microsoft Foundry Agents Playground



Demo: Develop, and Test Agents



Demo: **Deploy Your AI Agent**



Introduction to Coding AI Agents





Introduction to Coding AI Agents

- Use AI Foundry to generate code for model calls, agents
- Introduction to Semantic Kernel
- Introduction to AutoGen
- Introduction to Microsoft Agent Framework



Poll: Which tool is currently in public preview?

- Microsoft Foundry
- Microsoft Agent Framework
- AutoGen
- Semantic Kernel



Use AI Foundry to Generate Code for Model Calls, Agent



- Microsoft Foundry
 - a. Generates code snippets
 - b. Lets you build AI agents and integrate the latest AI models into your code
 - i. C#
 - ii. Python
 - iii. REST API
 - iv. ...



Developing Agents with Semantic Kernel

- Semantic Kernel
 - a. Single agents, tool/plugin orchestration, enterprise integration
 - b. Develop agents in other platforms, other than Azure
 - c. Build AI agents and integrate the latest AI models into your code
 - i. C#
 - ii. Python
 - iii. Java
 - d. Can add your existing code as a [plugin](#)



Developing Agents with AutoGen

- [AutoGen](#)
 - a. Multi-agent system, complex conversations, experimentation
 - b. Develop agents in other platforms, other than Azure
 - c. Open-Source Framework for Agentic AI
 - d. [QuickStart](#)



Microsoft Agent Framework

- a. Microsoft Agent Framework is a framework for building AI agents
 - a. That can reason, use tools, access data, and act autonomously
 - b. Standardizes agent design, orchestration, and tool use
 - c. Why?
 - a. Microsoft: *“Microsoft Agent Framework is now the successor path for both AutoGen and Semantic Kernel. Microsoft describes it as combining AutoGen’s simple agent abstractions with Semantic Kernel’s enterprise features like state management, type safety, middleware/filters, telemetry, model support, and explicit workflow orchestration.”*



Demo:

Create a New Agent (C#)



Demo:

Create a New Agent (Python)



Demo: **Semantic Kernel**



Demo: AutoGen



Demo:

Microsoft Agent Framework



Introduction to No-code Agent Development with Copilot Studio





Design and Implement Agents in Microsoft Copilot Studio

- Understanding Microsoft Copilot Studio



Poll: Which tool allows you to build and deploy AI agents without any coding skills?

- Azure AI Foundry
- Microsoft Copilot Studio
- AutoGen
- Semantic Kernel



Developing Agents with Microsoft Copilot Studio

- Copilot Studio
 - a. Copilot Studio is a graphical, low-code tool for building AI agents!
 - b. Can connect to other data sources (RAG)
 - c. Copilot Studio
 - i. Web App at <https://copilotstudio.microsoft.com>
 - ii. Teams app



Developing Agents with Microsoft Copilot Studio

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Conclusion





Resources

- [Azure AI Foundry](#)
- [Upgrade from Azure OpenAI to Azure AI Foundry](#)
- [Azure OpenAI in Azure AI Foundry Models quotas and limits](#)
- [Add and configure models to Azure AI Foundry Models](#)
- [What is Azure OpenAI in Azure AI Foundry Models?](#)
- [Azure AI Foundry Playgrounds](#)
- [What is Azure AI Foundry Agent Service?](#)
- [Set up your environment \(AI Foundry\)](#)
- [Quickstart: Create a new agent \(Python, AI Foundry\)](#)
- [Quickstart: Create a new agent \(C#, AI Foundry\)](#)
- [Models supported by Azure AI Foundry Agent Service](#)
- [What are tools in Azure AI Foundry Agent Service?](#)
- [Azure AI Foundry Agent Service quotas and limits](#)
- [Azure AI Foundry pricing](#)
- [Pricing calculator](#)
- [What's new in Azure AI Foundry Agent Service](#)
- [Threads, Runs, and Messages in Azure AI Foundry Agent Service](#)
- [Built-in enterprise readiness with standard agent setup](#)
- [Build collaborative, multi-agent systems with Connected Agents](#)



Resources

- [Evaluate your AI agents locally with the Azure AI Evaluation SDK \(preview\)](#)
- [Monitor Azure AI Foundry Agent Service](#)
- [Develop an AI agent with Azure AI Foundry Agent Service \(training module\)](#)
- [Call Azure Logic apps as functions using Azure OpenAI Assistants](#)
- [Trigger an agent using Logic Apps](#)
- [Azure OpenAI Assistants function calling](#)
- [Azure AI Foundry SDK client libraries](#)
- [Retrieval Augmented Generation \(RAG\) in Azure AI Search](#)
- [Model Context Protocol \(MCP\)](#)
- [Azure MCP Server](#)
- [Copilot Studio overview](#)
- [Add knowledge to an agent](#)
- [Semantic Kernel](#)
- [AutoGen](#)
- [Microsoft Agent Framework](#)
- [AI agent adoption](#)
- [Technology plan for AI agents](#)



CERTIFICATION

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Demonstrate fundamental AI concepts related to the development of software and services of Microsoft Azure to create AI solutions.

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 **Role**

 **Subject**



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AI Builder

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Artificial intelligence



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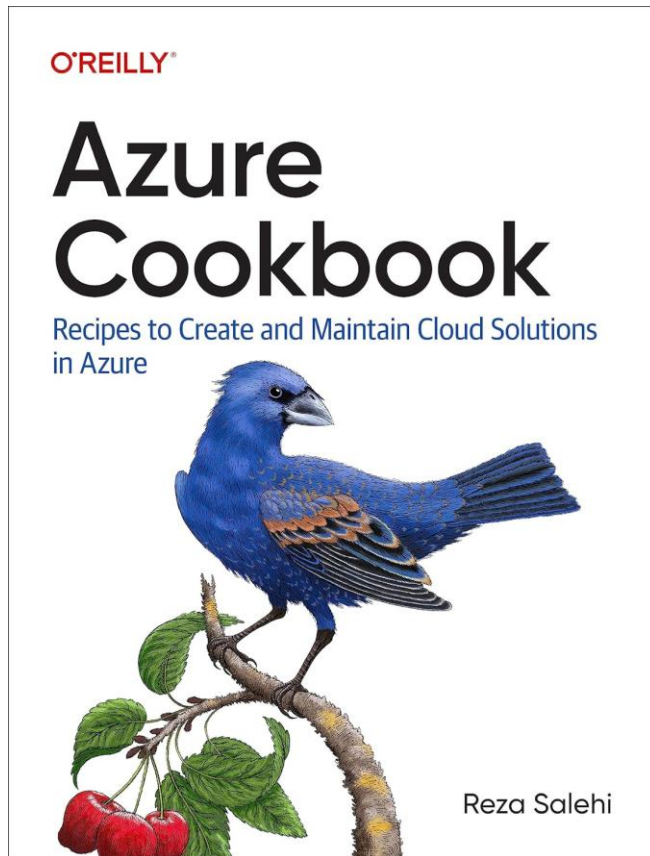
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Azure Cookbook

- <https://learning.oreilly.com/library/view/azure-cookbook/9781098135782/>
- <https://www.amazon.ca/Azure-Cookbook-Recipes-Maintain-Solutions/dp/1098135792/>
- <https://www.amazon.com/Azure-Cookbook-Recipes-Maintain-Solutions/dp/1098135792>



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